MSc. in Computing

Practicum Approval Form

Section 1: Student Details

Project Title:	Retinal Disease Detection Using Deep Learning
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Chosen major:	Data Analytics
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Date of Submission	17 November 2025

Section 2: About your Practicum

Please answer all questions below. Please pay special attention to the word counts in all cases.

What is the topic of your proposed practicum? (100 words)

The proposed practicum focuses on developing an automated retinal disease detection system using deep learning and Explainable AI (XAI). The project aims to classify fundus images into categories such as diabetic retinopathy, Age-related Macular Degeneration, Macular Hole, Yellow deposits under the retina, Diabetic Macular Edema, Central Serous Retinopathy, Choroidal Neovascularization and normal. XAI will be implemented to generate clear visual explanations that highlight the specific retinal regions influencing each prediction and translate these findings into simple medical terms. This ensures transparency, clinical interpretability, and greater

trust in the system. The final solution supports early screening and is suitable for telemedicine and primary-care environments where specialist review is limited.

Please provide details of the papers you have read on this topic (details of 5 papers expected).

1. M. Mariam and R. Thakur, “Optimizing Retinal Disease Detection: A Comparative Analysis of State-of-the-Art Deep Learning Models for Multiclass Classification”.

Link: Optimizing Retinal Disease Detection: A Comparative Analysis of State-of-the-Art Deep Learning Models for Multiclass Classification | IEEE Conference Publication | IEEE Xplore

2. K. I. Pattan, M. A. Aman, R. J. Patil, R. Mohamad, and P. S. P., “Multi-Disease Detection Using Hybrid Models and Transfer Learning: Cardiovascular, Pulmonary, Retinal and Renal”.

Link: Multi-Disease Detection Using Hybrid Models and Transfer Learning: Cardiovascular, Pulmonary, Retinal and Renal | IEEE Conference Publication | IEEE Xplore

3. K. C. Pavithra, P. Kumar, M. Geetha, S. V. Bhandary, K. B. A. Shenoy, G. Rao, S. Fernandes, and A. Tulsani, “Transformer-Based DME Classification Using Retinal OCT Images Without Data Augmentation: An Evaluation of ViT-B16 and ViT-B32 With Optimizer Impact,”

Link: Transformer-Based DME Classification Using Retinal OCT Images Without Data Augmentation: An Evaluation of ViT-B16 and ViT-B32 With Optimizer Impact | IEEE Journal Publication | IEEE Xplore

4. S. R. Gnanamalar, S. V. Prasanth, K. Sanjay, N. Sanjay, and S. Varnika, “Retinal Disease Detection and Diagnosis Using Hybrid ResNet-RNN Algorithm”.

Link: Retinal Disease Detection and Diagnosis Using Hybrid ResNet-RNN Algorithm | IEEE Conference Publication | IEEE Xplore

4. P. Ramya, S. Shona, S. Varshini, S. Vinothini, and Y. Sumjana, “Detection of Multiple Retinal Diseases using a Pretrained Convolutional Neural Network Model”.

Link: Detection of Multiple Retinal Diseases using a Pretrained Convolutional Neural Network Model | IEEE Conference Publication | IEEE Xplore

5. L. B. Latha, V. Sindhuja, B. Vinoth Kumar, K. Dhanush Kumar, S. Kavipriya, and S. Kiruthika, “Detection of Retinal Diseases Using Deep Learning Approaches”.

Link: Optimized CNN for Multi-Class Retinal Disease Detection with Minimized Memory Usage | IEEE Conference Publication | IEEE Xplore

How does your proposal relate to existing work on this topic described in these papers? (200 words)

Existing studies on retinal disease detection primarily focus on improving classification accuracy using CNN and hybrid deep learning models. While many of these papers successfully classify multiple retinal diseases, most rely on a single explainability technique typically Grad-CAM and provide only visual heatmaps without deeper clinical interpretation. Our work extends this by introducing a more comprehensive and practical explainability framework.

The novelty of our proposal is the integration and comparison of multiple XAI methods such as Grad-CAM, Grad-CAM++, and Eigen-CAM. By analysing their strengths and weaknesses, we aim to determine which method produces the clearest and most clinically relevant explanations. This comparative approach is largely missing in the existing literature.

Additionally, unlike prior works that stop at visual heatmaps, our project introduces a rule-based explanation module that converts heatmap regions into simple, human-understandable diagnostic statements. Conventional heatmaps offer only rough visual cues and cannot support clinical decision-making, as they do not convey region importance, pathology localization or confidence measures required by doctors. This bridges the gap between raw visual outputs and meaningful clinical interpretation, making the system more useful for clinicians and non-experts.

Thus, our proposal contributes a more transparent, interpretable, and clinically aligned framework that enhances trust and supports real-world adoption in early screening and telemedicine workflows going beyond the accuracy-centric focus seen in the reviewed papers.

What are the research questions that you will attempt to answer? (200 words)

This practicum aims to address several key research questions related to automated retinal disease detection using OCT imaging and Explainable AI.

First, to what extent can a deep learning–based model accurately classify multiple retinal diseases when trained on an eight-class OCT dataset containing conditions such as CNV, DME, Drusen, CSR, Macular Hole, RAO, RVO, and Normal? This question evaluates the feasibility of a unified multiclass diagnostic model.

Second, which retinal features or regions contribute most to the model's predictions, and can these be reliably visualised using Grad-CAM, Grad-CAM++, Eigen-CAM? This examines whether interpretability techniques can highlight clinically relevant structures within OCT layers. Third, can XAI explanations be translated into simple, medically meaningful descriptions that help users understand why a particular diagnosis was made? This question focuses on bridging the gap between technical outputs and clinical interpretability.

Finally, how well does the model generalize to variations in image quality, noise, and structural differences across disease classes? Answering this assesses the robustness and real-world applicability of the proposed system. Together, these research questions guide the development of a transparent, clinically trustworthy, and scalable retinal disease detection framework.

How will you explore these questions? (Please address the following points. Note that three or four sentences on each will suffice.)

- **What software and programming environment will you use?**

The practicum will be developed primarily in Python, using deep learning and explainability frameworks suitable for medical image analysis. Model training and evaluation will be carried out in environments such as Jupyter Notebook, Google Colab, or VS Code. The core classification model will be implemented using Convolutional Neural Networks (CNNs) built with TensorFlow and Keras. Image

preprocessing, augmentation and OCT data handling will be performed using libraries such as OpenCV, NumPy, and Pandas.

To support interpretability, Grad-CAM, Grad CAM++, Eigen CAM will be integrated to generate visual heatmaps that identify the retinal regions influencing each prediction and compare the model's performance. For explainability beyond visualization, an XAI Report Agent will be implemented in Python to convert heatmap findings into simple, clinically meaningful descriptions. Additional libraries such as Matplotlib and Seaborn will be used for visualization and result analysis. The entire workflow will run on GPU-supported environments when available to accelerate training.

- **What coding/development will you do?**

The development work for this practicum will involve building a complete end-to-end retinal disease detection and explainability pipeline. First, I will implement Python scripts for preprocessing OCT images, including resizing, normalization, augmentation, and dataset structuring. Next, I will develop and train a Convolutional Neural Network (CNN) model for multiclass classification of the eight retinal diseases in the OCT C8 dataset. This includes writing code for model architecture selection, training loops, validation routines, performance monitoring, and hyperparameter tuning.

To support explainability, I will integrate Grad-CAM, Grad-CAM++, Eigen-CAM into the system by coding methods that extract gradients, feature maps and generate visual heatmaps showing which retinal regions influenced each prediction. Beyond visualization, I will build an XAI Report Agent in Python to interpret the Grad-CAM outputs and automatically produce simple medical-style explanations that describe why a particular disease was predicted. The XAI Report Agent uses a template-based explanation generator that converts numerical features extracted from heatmaps (such as region score, centroid, and predicted class) into structured, human-readable statements. This makes the explanations consistent and reliable, because everything is generated from fixed templates using only the values produced by the model.

Additional coding will include implementing evaluation metrics, confusion matrices, plotting tools, and exporting results for analysis. The final development stage will involve creating a structured pipeline that processes input images, performs classification, generates explanations and outputs a combined diagnostic report.

- **What data will be used for your investigations?**

The dataset contains Optical Coherence Tomography (OCT) images representing eight retinal conditions, making it suitable for multiclass disease classification and explainability research. The classes included are:

- CNV (Choroidal Neovascularization)
- DME (Diabetic Macular Edema)
- Drusen
- CSR (Central Serous Retinopathy)
- Macular Hole (MH)
- Retinal Artery Occlusion (RAO)
- Retinal Vein Occlusion (RVO)
- Normal

The dataset provides high-resolution OCT slices that capture structural retinal abnormalities at layer level, allowing CNN models and Grad-CAM, Grad-CAM++, Eigen-CAM explainability methods to identify clinically meaningful patterns. It also includes sufficient variability across disease categories to support training, validation, and testing splits. Referenced MedMNIST v2, which includes the OCTMNIST subset a widely used retinal OCT benchmark commonly employed for evaluating classification and explainability methods.

This dataset is appropriate for developing and evaluating a deep learning-based diagnostic system and for generating XAI-based explanations due to its diversity, clear class labels, and real clinical imaging characteristics.

- Is this data currently available, it not, where will it come from?

Yes, the data required for this practicum is already available. The project will use the publicly accessible Retinal OCT C8 dataset, available on Kaggle, which can be downloaded directly without restrictions for academic research purposes.

Dataset Link: https://www.kaggle.com/datasets/obulisainaren/retinal-oct-c8

- What experiments do you expect to run?

Experiments will include training the classification model with and without augmentation, comparing CNNs with enhanced pre-trained models, and testing robustness against noisy or low-quality images. Multiple XAI methods (Grad-CAM, Grad-CAM++, and Eigen-CAM) will be applied to generate comparative heatmaps. The XAI Report Agent will be tested for producing correct and meaningful diagnostic explanations. Model performance will be evaluated using accuracy, precision, recall, F1-score, confusion matrices, and interpretability assessments.

-What output do you expect to gather?

Expected outputs include a trained multiclass classification model with comprehensive performance metrics, visual heatmaps from multiple XAI methods, a comparison of interpretability quality, and human-readable diagnostic explanations generated by the XAI Report Agent. These outputs will demonstrate both predictive accuracy and clinical interpretability.

- How will the results be evaluated?

Evaluation will include accuracy, precision, recall, and F1-scores across all eight disease classes, supported by confusion matrices. Heatmaps generated by Grad-CAM, OI, and Eigen-CAM will be reviewed for clinical relevance and compared for interpretability quality. The rule-based XAI explanation module will be assessed for clarity, correctness, and usefulness to non-experts. Robustness will be evaluated by

testing on varied image conditions and analysing consistency across validation and test sets.